

JIAJIE (JERRY) LUO

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Last update: March 23, 2026

EMPLOYMENT

Postdoctoral Scholar

October 2024 – Present

The Knowledge Lab

University of Chicago

Faculty Mentor: Professor James Evans

Ph.D. Research Intern

June 2022 – September 2022

Mathematics, Statistics, and Data Science

Pacific Northwest National Laboratory

Mentors: Dr. Tegan Emerson; Dr. Gregory Henselman-Petrusek Roek

EDUCATION

University of California, Los Angeles

September 2019 – June 2024

Ph.D. in Mathematics.

Thesis Title: Topics in Persistent Homology and Complex Social Systems

Advisor: Professor Mason Porter

University of California, Santa Barbara

September 2017 – June 2019

M.A. in Mathematics.

Thesis Title: On Abstract Witt Rings and Quadratic Extensions

Advisor: Professor Bill Jacob

University of California, Santa Barbara

September 2014 – June 2017

College of Creative Studies

B.S. in Mathematics, *Highest Honors*

Faculty Advisor: Professor Jeffrey Stopple

RESEARCH INTERESTS

Topological Data Analysis, Persistent Homology and Applications, Complex Systems, Opinion Dynamics on Networks

PREPRINTS & PUBLICATIONS

1. **J. Luo**, G. Henselman-Petrusek, *Interval Decomposition of Infinite Persistence Modules over a Principal Ideal Domain*, To Appear in Proceedings of the American Mathematical Society, arXiv:2511.07614
2. G. J. Li, **J. Luo**, W. Chu, *Bounded-Confidence Models of Multi-Dimensional Opinions with Topic-Weighted Discordance*, Published in *SIAM Journal on Applied Dynamical Systems*.
3. **J. Luo**, G. Henselman-Petrusek, *Interval Decomposition for Persistence Modules Over a Principal Ideal Domain*, Published in *Foundations of Computational Mathematics*.

4. G. J. Li*, **J. Luo***, M. A. Porter (*Equal Contribution), *Bounded-Confidence Models of Opinion Dynamics with Adaptive Confidence Bounds*, Published in *SIAM Journal on Applied Dynamical Systems*.
5. A. Hickok*, B. Jarman*, M. C. Johnson*, **J. Luo***, M. A. Porter (*Equal Contribution), *Persistent Homology for Resource Coverage: A Case Study of Access to Polling Sites*, Published in *SIAM Review*.
6. V. Chayes, K. Miller, R. Bhalerao, **J. Luo**, W. Zhu, A. Bertozzi, W. Liao, S. Osher, *Pre-Processing and Classification of Hyperspectral Imagery Via Selective Inpainting*, Published in *ICASSP2017*.

EXPOSITORY ARTICLES

G. J. Li, **J. Luo**, K. Peng, and M. A. Porter. *Using Mathematics to Study How People Influence Each Other's Opinions*, Published in *Frontiers for Young Minds*.

AWARDS, HONORS & FELLOWSHIPS

Pacific Journal of Mathematics Dissertation Prize	2024
ModELing and uNdersTanding human behaviOR (MENTOR) Fellowship	2021–2022
College of Creative Studies Commencement Speaker	2017
Adil Yaquub is my Hero Scholarship	2016

TALKS & PRESENTATION

IMSI Workshop: The Geometric Realization of AATRN	August 2025
Interval Decomposition of Persistence Modules over a Principal Ideal Domain (Poster + Lightning Talk)	
IMSI Workshop: Emergent Behavior in Complex Systems of Interacting Agents	March 2025
Bounded-Confidence Models of Opinion Dynamics with Adaptive Confidence Bounds (Poster Session)	
Southern California Applied Mathematics Symposium (SOCAMS)	April 2024
Bounded-Confidence Models of Opinion Dynamics with Adaptive Confidence Bounds	
Graduate Student Topology and Geometry Conference (GSTGC2024)	April 2024
Interval Decomposition of Persistence Modules over a Principal Ideal Domain (Poster Session)	
Joint Mathematics Meetings 2024 (JMM 2024)	January 2024
AMS Special Session on Complex Social Systems I	
Persistent Homology for Assessing Facility Placement (Invited Talk)	
2023 Algorithms for Threat Detection PI Workshop (ATD2023)	October 2023
Bounded-Confidence Models of Opinion Dynamics with Adaptive Confidence Bounds	
Computation Persistence Workshop (ComPer23)	September 2023
Interval Decomposition for Persistence Modules of Free Abelian Groups	
SIAM Conference on Applications of Dynamical Systems (DS23)	May 2023
Bounded-Confidence Models of Opinion Dynamics with Adaptive Confidence Bounds	
Southern California Applied Mathematics Symposium (SOCAMS)	April 2023
Persistent Homology for Resource Coverage: A Case Study of Access to Polling Sites	
SIAM Workshop on Network Science (NS22)	September 2022
Bounded-Confidence Models with Adaptive Confidence Bounds	
Virtual Research Symposium, Pacific Northwest National Laboratory.	August 2022
Topological Data Analysis and Machine Learning	

TEACHING EXPERIENCE

As Graduate Student Instructor (UCLA)

Math 110A: Abstract Algebra Winter 2024

Math 115A: Linear Algebra (proof-based) Winter 2023

As Graduate Teaching Assistant (UCLA)

Math 31AL: Differential and Integral Calculus Laboratory Winter 2021

Math 115A: Linear Algebra (proof-based) Fall 2020, Spring 2021

Math 31B: Integration and Infinite Series Spring 2020

Math 33A: Linear Algebra and Application Winter 2020, Fall 2020, Spring 2021

Math 3B: Calculus for Life Sciences II Fall 2019, Winter 2021

As Graduate Teaching Assistant (UCSB)

Math 117: Methods of Analysis Spring 2019

Math 108A: Introduction to Linear algebra (proof-based) Winter 2019

Math 4A: Linear Algebra and Applications Fall 2018

MATH 100B: Mathematics for Elementary Teaching II Summer 2018

Math 34B: Calculus for Social Sciences II Winter 2018, Spring 2018

Math 34A: Calculus for Social Sciences I Fall 2017

UNDERGRADUATE MENTORING

Research Mentoring:

William Flowers — Bounded-Confidence Models of Opinion Dynamics Fall 2024 – Present

Yuxuan Wu — A Bounded-Confidence Model with Adaptive Edge Weights Summer 2024 – Present

Leila Thompsky — A Bounded-Confidence Model with Adaptive Edge Weights Fall 2023 – Present

Amos Ancell — Persistent Homology for Resource Coverage Fall 2023 – Spring 2024

Ruyi Lu — Bounded-Confidence Models on Random Configuration Models Winter 2023 – Fall 2023

Xinyue (Serena) Li — Persistent Homology for Resource Coverage Winter 2023 – Spring 2023

Xiaohe (Haley) Zhang — Bounded-Confidence Models with Repulsion Winter 2022 – Spring 2022

Directed Reading Program:

DRP Committee Fall 2021 – Spring 2024

Students:

Yuxuan (Yolanda) Wu — Models of Opinion Dynamics Spring 2024

Leila Thompsky — Complex Social Systems Fall 2023

Amos Ancell — Applied Topology, Persistent Homology Winter 2023 – Spring 2023

Xinyue (Serena) Li — Applied Topology, Persistent Homology Fall 2022 – Winter 2023

Ruyi Lu — Opinion Dynamics on Networks Fall 2022 – Winter 2023

Haoyang Lyu — Applied Topology, Persistent Homology Winter 2022 – Spring 2022

Chenxin (Amy) Shen — Applied Topology, Persistent Homology Fall 2021 – Spring 2022

Xiaohe (Haley) Zhang — Opinion Dynamics on Networks Fall 2021 – Winter 2022

Tanishq Bhatia — Topics in Persistent Homology Winter 2021 – Spring 2021

Other Mentoring:

Mentor for UCLA Applied Mathematics REU (ATD Traffic Challenge) Summer 2021

Students:

- Matthew Hudes (Tufts University)
- Naji Sarsam (UCLA)
- Chenxin (Amy) Shen (UCLA)
- Wenwen Tang (USC)

MISCELLANEOUS

Citizenship: United States

Programming Experience: Python, MATLAB, R., C++

Languages: Chinese (Mandarin), English.